Jinhyeok Park

Email: jpark3263@gatech.edu | LinkedIn: linkedin.com/in/jinhyeok-park | Website: jpark3263.github.io

OBJECTIVE

Ph.D. student in ECE working across mmWave/RF hardware, real-time radar system, and efficient ML (state-space models).

EDUCATION

Ph.D., Electrical and Computer Engineering, Georgia Institute of Technology, USA *Aug 2024 – Present*
GPA: 3.85/4.0 | Expected Graduation: Aug 2028 | Advisor: Prof. Saibal Mukhopadhyay

M.S., Electrical Engineering, Korea Advanced Institute of Science and Technology (KAIST), South Korea *Mar 2021 – Feb 2023*

B.S., Electrical Engineering, Sungkyunkwan University, South Korea *Mar 2014 – Feb 2021*

RESEARCH EXPERIENCE

Graduate Research Assistant (PhD), GREEN Lab, Georgia Tech *Aug 2024 – Present*

- **Anytime NLOS radar imaging**: complex-valued selective SSM that reconstructs hidden objects from streaming mmWave frames, matching the offline state of the art (F-score 0.70 vs 0.73) at 10x lower end-to-end time.
- **Physics-informed Kalman filtering**: Kalman filtering recast as a parallel selective SSM; +5.5dB accuracy, +9.5dB under frame drops, 2.5x faster inference than neural Kalman filters.
- **Real-time adaptive radar**: per-frame RF front-end/ADC tuning on AWR1843; 33-69% lower total energy at 21.76ms latency
- **EM side-channel monitoring of AI workloads**: Building antenna/SDR sensing and signal processing to identify GPU workloads from their EM emissions.

Graduate Teaching Assistant, ECE 6130 Advanced VLSI, Georgia Tech *Fall 2026*

Graduate Research Assistant (MS), WEIS Lab, KAIST *Mar 2021 – Feb 2023*

- Designed a 23–35 GHz wideband bidirectional VGA and 24-30 GHz power amplifier for 5G beamforming ICs.

INDUSTRY EXPERIENCE

RF Engineer, Samsung Electronics, South Korea *Mar 2023 – Apr 2024*

- Developed 60 GHz FMCW Radar built-in-self-test modules for mobile sensing applications.
- Designed and validated 28/39 GHz phased-array RF transceiver modules for commercial 5G products

OPEN-SOURCE TOOLS

- **mmwcas-stream**: live raw-ADC streaming from TI's 4-chip cascade radar (12TX / 16RX) to a PC at ~4ms latency.
- **RT-FEA**: real-time RF front-end adaptation loop for TI AWR1843BOOST with DCA1000

PUBLICATIONS

[1] **J. Park** et al., "Anytime NLOS Radar Imaging with Complex-Valued Selective State-Space Models" *WACV* 2027.

[2] **J. Park** et al., "Range-Adaptive FMCW Radar: Real-Time Adaptation of RF and Baseband for Robust and Energy-Efficient Sensing," *IMS RFSA* 2026.

[3] **J. Park** et al., "Parallel Kalman Filtering with Physics-Informed Selective State-Space Models for Robust Radar Sensing," *IMS RFSA* 2026.

[4] **J. Park** et al., "Wideband Bidirectional Variable Gain Amplifier for 5G Communication," *IEEE MWTL*, 2023.

[5] S. Sharma, **J. Park** et al., "LUGA: Lightweight Uncertainty-Guided Sensing Resolution Adaptation for Energy Efficient Radar Processing," *IEEE SENSORS 2025*.

[6] **J. Park** et al., "Ego-Sovereign Fusion: Risk-Controlled Preservation for Cooperative Perception" (ICRA 2027 under review)

[7] H. Cho, **J. Park** et al, "State as Messages: Latent Communication for State Space Models" (ICLR 2027 under review)

[8] A. Padhy, **J. Park** et al, "ReBound: Readout-Aware Boundary Pruning for Efficient Mamba Prefill" (ICLR 2027 under review)

TECHNICAL SKILLS

ML / Sequence modeling: state-space models, physics-informed learning, Kalman filtering, conformal prediction
Signal Processing: FMCW, Range-Doppler/Angle, CFAR, Capon beamforming, SAR/coherent integration, waveform design
Hardware & Embedded: TI mmWave sensor (AWR1843, MMWCAS), SDR, FPGA (RFSoC), Cadence Virtuoso, ADS
Programming: Python, PyTorch, MATLAB, C/C++

AWARDS AND HONORS

Georgia Tech Cironnet Fellowship, for contribution to education (2026)
Georgia Tech ECE Fellowship (2024-2025)
Best Poster Award, SRC Cognisense Annual Review (2025)
Samsung Semiconductor Scholarship (2021-2023)